Impulse Control: Trading in CEXs and DEXs with Priority Fees and Stochastic Delays

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CEX, DEX, Key Notations

The Problem:

- An agent trades between a Centralized Exchange (CEX) and a Decentralized Exchange (DEX).
- **CEX:** Continuous trading with immediate execution.
- **DEX:** Discrete orders (impulses) with **stochastic execution delays**.
- Agent can influence the **mean delay** by choosing a **priority fee** $\mathfrak{p}_i$.

Key Mathematical Objects:

- **Delay:** $\ell_i > 0$, the intensity of the Poisson process for priority level i (higher fee $\rightarrow$ higher ℓ_i $\rightarrow$ shorter expected delay).
- **Impulse Control:** $(\tau_n, I_n, \xi_n)_{n \geq 1}$ where:
  - τ_n: Intervention time (order submission).
  - $I_n \in \mathcal{I} = [1, N]$: Priority fee index ($p_{I_n}$)
  - $\xi_n \in \mathcal{U} = [-\hat{V}, \hat{V}]$: Order size (volume).
- **Execution Time:** $\tilde{\tau}_n$, the first jump of the Poisson process N^{I_n} after τ_n. $\tilde{\tau}_n - \tau_n \sim \text{Exp}(\ell_{I_n})$.

State Variables and Pending Orders

Pending Orders:

- $k(t,\alpha) = \text{card}(\iota(t,\alpha)) \leq K$: Number of pending orders.
- $\mathbf{i}(t,\alpha) = (i_1, \ldots, i_N)$: Count of pending orders per priority level.
- $\mathbf{v}(t,\alpha) = (v_1, \ldots, v_N)$: Total pending volume per priority level.
- $p(t,\alpha) = (\mathbf{i}(t,\alpha), \mathbf{v}(t,\alpha))$: The state of pending orders.

Example: $K = 3$, orders: #2 (lvl1,+10), #5 (lvl1,-3), #7 (lvl3,+7)

$$\iota(t,\alpha) = \{2,5,7\},\ \mathbf{i}(t,\alpha) = (2,0,1),\ \mathbf{v}(t,\alpha) = (7,0,7)$$

Admissible Strategies $\mathcal{A}_K$: Set of controls $\alpha = ((\nu_t)_{t\in[0,T]}, (\tau_n, l_n, \xi_n)_{n\geq 1})$ such that $k(t,\alpha) \leq K$. **Value Function:**

$$v(t,x,\mathbf{i},\mathbf{v}) = \sup_{\alpha\in\mathcal{A}_{K,\mathbf{i},\mathbf{v}}(t)} J(t,x,\alpha)$$

where x is the state vector (e.g., prices, inventory).

Model Setup: State Dynamics

Controlled State: $X^\alpha = (S^\alpha, Q^\alpha, Z^\alpha)$.

- **CEX Price (S):** External price.

$$S_u^{t,s,\alpha} = s + \sigma^S (W_u^S - W_t^S) \tag{16}$$

- **DEX Price (Z):** Mean-reverts to CEX price.

$$\begin{aligned} Z_u^{t,z,s,\alpha} = z + \int_t^u \kappa(S_r^{t,s,\alpha} - Z_r^{t,z,s,\alpha})dr + \sigma^Z (W_u^Z - W_t^Z) \\ + \sum_{\tilde{\tau}_n \leq u} h(\xi_n, Z_{\tilde{\tau}_n^-}^{t,z,s,\alpha}), \quad h(\xi, Z) = \psi(Z)\xi \end{aligned} \tag{17}$$

- **Inventory (Q):**

$$Q_u^\alpha = q + \underbrace{\int_t^u \nu_r dr}_{\text{CEX trading}} + \underbrace{\sum_{n:\tilde{\tau}_n \leq u} \xi_n}_{\text{DEX trades}} \tag{18}$$

Model Setup: Rewards and Costs

Cash and Performance:

- **Cash (C):**

$$C_u^\alpha = -\underbrace{\int_t^u \nu_r(S_r^\alpha + k\nu_r)dr}_{\text{CEX proceeds}} + \underbrace{\sum_{n:\tilde{\tau}_n \le u} \left(\gamma(\xi_n, Z_{\tilde{\tau}_n^-}^\alpha) - \mathfrak{p}_{I_n}\right)}_{\text{DEX proceeds net of fees}} \tag{19}$$

- **Intervention Cost c:** Net cost of a DEX trade.

$$c(\tilde{\tau}_n, Z_{\tilde{\tau}_n^-}^\alpha, \xi_n, I_n) = \gamma(\xi_n, Z_{\tilde{\tau}_n^-}^\alpha) - \mathfrak{p}_{I_n} \text{ where } \gamma(\xi, z) = \frac{d^2}{\xi - d\sqrt{1/z}} + d\sqrt{z}.$$

- **Running Reward f:** Penalty for trading speed and inventory.

$$f(t, y, \nu) = -\nu(s + k\nu) - \phi q^2 \tag{21}$$

- **Terminal Payoff g:** Liquidation value.

$$g(y) = qs - \Xi q^2 \tag{21}$$

Model Setup: Rewards and Costs

- **Performance Criterion J:**

$$J(t,y,\alpha) = \mathbb{E}_{t,y}\Big[\int_t^T -\nu_r(S_r^\alpha + k\nu_r)dr + Q_T^\alpha S_T^\alpha - \Xi(Q_T^\alpha)^2 - \sum_{\tilde{\tau}_n \in (t,T]} c(\tilde{\tau}_n, Z_{\tilde{\tau}_n}^\alpha, \xi_n, I_n) - \phi\int_t^T (Q_r^\alpha)^2 dr\Big] \quad (22)$$

- **Value Function:**

$$v(t,y) = \sup_{\alpha \in \mathcal{A}_{K,\mathbf{i},\mathbf{v}}(t)} J(t,y,\alpha)$$

The HJB Quasi-Variational Inequality

The value function v is the viscosity solution of the following HJBQVI. **If number of pending orders $< K$:**

$$\min\left\{-\frac{\partial v}{\partial t}(t,y) - \sup_{\nu\in\mathbb{R}} H^{\nu}\left(t,y,v,\frac{\partial v}{\partial x},\frac{\partial^2 v}{\partial x^2}\right),\ (v-\mathcal{M}v)(t,y)\right\} = 0 \tag{23}$$

- **Hamiltonian H^{ν}:** Includes infinitesimal generator $\mathcal{L}^{\nu}$, execution of pending orders $\mathcal{J}$, and running rewards f.

$$H^{\nu}(\cdot) = \mathcal{L}^{\nu}v + \mathcal{J}v - \phi q^2 - \nu(s + k\nu)$$

- **Intervention Operator $\mathcal{M}$:** Represents the value of submitting a new order.

$$\mathcal{M}\phi(t,s,q,z,\mathbf{i},\mathbf{v}) = \sup_{(\xi,i)} \phi(t,s,q,z,\mathbf{i}+e_i,\mathbf{v}+\xi e_i) \tag{24}$$

- Optimal CEX trading speed: $\nu^* = (2k)^{-1}(\frac{\partial v}{\partial q} - s)$.

The HJB Quasi-Variational Inequality

If number of pending orders $= K$**:** The $\mathcal{M}$ term disappears (cannot submit new orders).

$$-\frac{\partial v}{\partial t} - \sup_{\nu \in \mathbb{R}} H^{\nu}(\cdot) = 0 \tag{25}$$

Numerical Results

Optimal trading strategy Figure 1 shows the value function and the corresponding exercise region for when the inventory is zero ($q = 0$).

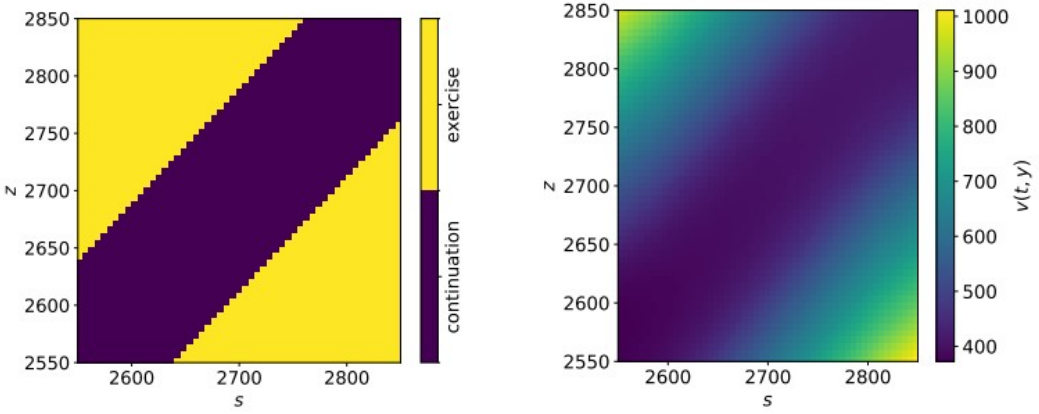
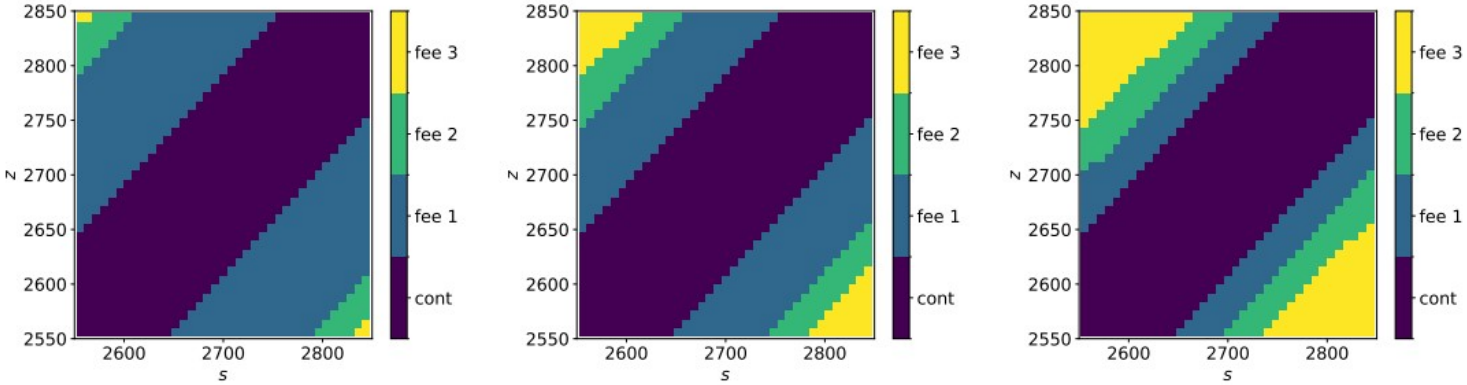


Figure 1: Value function and exercise region as a function of the CEX and DEX prices (s, z) for $q = 0$.



(a) $\mathfrak{l} = \mathfrak{l}_0$ (b) $\mathfrak{l} = \mathfrak{l}_1$ (c) $\mathfrak{l} = \mathfrak{l}_2$

Figure 5: Priority fee regions for increasing exogenous arrival intensities at the CEX and DEX at time $t = 0.5$. From left to right, all arrival intensities are jointly increased $\mathfrak{l}_2 < \mathfrak{l}_1 < \mathfrak{l}_0$.

Conclusion and Limits

Conclusion:

- Developed a novel mixed control framework with **controlled stochastic delays** and **multiple pending orders**.
- Proved that the value function is the unique viscosity solution of the corresponding HJBQVI.
- Applied to CEX-DEX trading, showing how to optimally choose priority fees to manage latency risk and price dislocations.
- Demonstrated significant economic value in strategic fee selection.

Limits and Future Work:

- **Simple Price Model:** $dS_t = \sigma^S dW_t$
- **Finite Priority Fees:** The discretization of fees to a finite set N is a simplification, though shown to be effective.
- **Poisson Delays:** Independant from price, market.
- **Strong hypothesis:** Linear impact on CEX: Price $= S_t + k\nu_t$.